

ADITYA RAJ

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Professional Summary

Full-stack engineer building production-grade web applications with TypeScript, React/Next.js, and Python. Experienced in developing local-first architectures, headless browser automation pipelines, and machine learning classification systems with Clean Architecture. Focused on shipping reliable, user-centric software with thoughtful system design.

Education

National Institute of Engineering

Bachelor of Engineering (BE) in Information Science and Engineering
CGPA: 8.5

2023 – 2027

Mysore, Karnataka

Projects

Tsugi(t) — Local-First Habit Tracker & Task Management PWA | Next.js 16, React 19, [Github](#) | [Live](#)
TypeScript, Dexie.js (IndexedDB), Firebase, PWA, Tailwind CSS

- Engineered a local-first Progressive Web App with zero-latency offline interactions via Dexie.js (IndexedDB) and Firebase Firestore cloud synchronization with manual and background sync.
- Implemented algorithmic streak calculation with 16-week retention heatmaps and analytics dashboards for habit consistency tracking.
- Built a read-only habit sharing system with cryptographically generated 6-character claim codes and an access-mirror security model for cross-user visibility.
- Designed customizable dark/light themes with 6 accent palettes and responsive glassmorphism UI components using Tailwind CSS.

GradeBuddy — Academic Analytics & Portal Automation Platform | Next.js, React 19, [Github](#) | [Live](#)
TypeScript, Puppeteer, Cheerio, Docker, Node.js

- Built a full-stack web platform that automates CIE mark retrieval from a university portal using authenticated headless browser sessions, computing required SEE scores and simulating SGPA outcomes.
- Engineered a Puppeteer-based automation pipeline with authenticated session management, combining headless navigation with parallel in-browser fetch() calls for fast data extraction (~15s for 8 courses).
- Designed a production Docker deployment on Render.com with optimized Chromium flags for constrained container environments (512MB RAM, 0.1 CPU).
- Implemented credit-weighted grade computation algorithms (CIE/SEE standardization, SGPA calculation) aligned with VTU evaluation policies.

RansomGuard.ai — ML-Powered Network Threat Classification System | Python, FastAPI, [Github](#)
XGBoost, Random Forest, Scikit-learn, JavaScript, Chart.js

- Architected a cybersecurity analytics platform using Clean Architecture (DDD) that classifies network traffic into 10 attack categories using the UNSW-NB15 dataset, with ransomware lifecycle phase mapping.
- Implemented 6 GoF design patterns (Strategy, Factory, Observer, Singleton, Facade, Template Method) to enable runtime model switching between Random Forest and XGBoost inference pipelines.
- Built a RESTful prediction API using FastAPI with a real-time analytics dashboard showing live classification metrics, model usage, and request statistics.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C, SQL

Web Technologies: React, Next.js, FastAPI, Node.js, Tailwind CSS, REST APIs

Databases & Tools: MySQL, Firebase/Firestore, IndexedDB, Docker, Git, Linux

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Achievements

- Secured a **merit-based branch change** from Mechanical Engineering to Information Science Engineering after ranking among the top students in the Mechanical Engineering cohort in first year of BE (**9.5 CGPA**).
- Top 10 Finalist**, ComedKares Ideathon, NIE (2024).